Toward AI-Assisted Ship Design Review: A Sample Manuscript

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Abstract

This document is a sample manuscript for trying out the PDF review tool. Select any sentence with the mouse and add a comment. Comments are stored inside the PDF as standard annotations, so they remain visible in Acrobat, macOS Preview and Zotero after you save the file.

1. Introduction

Ship design is an iterative process in which requirements, hull form, general arrangement and machinery are negotiated over many rounds of review. Each round produces documents that must be read carefully by several people. In a research group this is also how students learn: a draft goes to a supervisor, comes back with comments, and is revised.

Existing tools either require a commercial license, store documents on a third-party server, or make it hard to merge comments from several reviewers. The tool you are using now keeps the PDF entirely inside the browser. Nothing is uploaded.

The remainder of this sample is filler text so that there is enough to select. Try highlighting a phrase, adding a sticky note in the margin, replying to your own comment, and then saving the PDF with Cmd+S.

2. Method

We consider a review workflow with three roles: an author, one or more reviewers, and a person who consolidates the feedback. Each reviewer receives the same PDF, annotates it independently, and returns the annotated copy through the usual channels (Slack, Box or e-mail).

The consolidator opens one copy in the tool and imports the other copies. Comments with the same identity are merged, and the remaining comments are shown side by side with the reviewer name and a colour per reviewer. The result can be saved as a single PDF or exported as a Markdown list grouped by page.

3. Results

Table 1 summarises the annotation types that survive a round trip through the tool. Highlight and sticky-note comments, including replies, keep their author, timestamp and text. Other annotation types present in the file are left untouched.

Type	Written	Read back	Notes
Highlight	yes	yes	QuadPoints + Popup
Text (note)	yes	yes	Icon: Comment
Reply	yes	yes	IRT / RT R
Ink, FreeText	no	counted	preserved as-is

4. Discussion

Because the geometry of every comment is stored in PDF user space, comments keep their position when the page is rotated or when the viewer zooms. The next page is rotated by 90 degrees to demonstrate this.

5. A rotated page

This page carries a /Rotate 90 entry. The text below is drawn so that it reads upright in a viewer. Highlight a few words here to confirm that the highlight lands on the right place after saving and reopening.

Acknowledgements. This sample was generated by scripts/make-sample.mjs using @cantoo/pdf-lib.